Lab Assignment | Network Utilities and Concepts

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This lab will cover essential networking utilities and provide you with hands-on experience in retrieving, interpreting, and analyzing network information. Follow the instructions provided for each task. Submit your comprehensive lab report in Canvas, ensuring it includes each item listed under the "Lab Report Requirements".

Task 1: Using IPCONFIG

Objective: To understand how to retrieve and interpret network configuration information using IPCONFIG.

Instructions

1. Open a command prompt on your computer (Windows users can type "cmd" in the search bar, and Unix/Linux users can open a shell, or search “terminal”).
2. In the command prompt or shell, type "ipconfig" and press Enter.
3. Record the following information from the IPCONFIG output.
    - Your computer's IP address

      192.168.56.1
    - Default gateway

      10.246.191.254
    - Subnet mask

      255.255.255.0
4. Provide a brief explanation of the significance of each piece of information in network configuration and troubleshooting.

   IP address is how data gets sent back from other devices

   Default Gateway is like the path you take to get into the internet

   Subnet Mask is determines if another device is on the same local network or remote network.

Lab Report Requirements

- The results and information obtained from the IPCONFIG command. (15 pt)

```
C:\Users\dylxn>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::e742:a335:4caf:15f8%18
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : vcu.edu
   Link-local IPv6 Address . . . . . : fe80::5ca9:b9ab:7b1b:8ec0%6
   IPv4 Address. . . . . . . . . . . : 10.246.144.94
   Subnet Mask . . . . . . . . . . . : 255.255.192.0
   Default Gateway . . . . . . . . . : 10.246.191.254

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

- Your analysis and reflections are based on the task. (15 pt)

Task 2: Using TRACERT

Objective: To gain insights into network routing by using the TRACERT utility.

Instructions:

1. Open a command prompt on your computer (Windows users can type "cmd" in the search bar, and Unix/Linux users can open a shell).
2. In the command prompt or shell, type "tracert [destination]" where [destination] is a domain or IP address of your choice (e.g., "tracert www.google.com")
3. Examine the output and identify the intermediate steps (hops) taken to reach the destination.
4. Perform this task for at least two different destinations.
5. Compare the hops taken for different destinations and analyze similarities and differences.

Both traces share identical hops 1–5 and both exit through Internet2 at hop 8, since that's VCU's fixed upstream provider regardless of destination.

YouTube took 28 hops with 12+ timed-out hops in the middle, while stake.us reached its destination in just 12 hops with almost no timeouts.

Lab Report Requirements

- The output and findings from the TRACERT utility. (15 pt)

```
C:\Users\dylxn>tracert youtube.com

Tracing route to youtube.com [142.251.163.91]
over a maximum of 30 hops:

  1    10 ms    34 ms     6 ms  10.246.191.254
  2     *        *        *     Request timed out.
  3     6 ms     2 ms     2 ms  ti-pe-default-eth2-11.ns.vcu.edu [172.16.4.25]
  4     6 ms     2 ms     2 ms  borderfw-wifi-out-3404.ns.vcu.edu [172.30.0.65]
  5     6 ms     3 ms     2 ms  border-rtr-tocb-hun1-250.ns.vcu.edu [192.168.250.2]
  6     9 ms     8 ms     9 ms  192.70.138.53
  7    10 ms     9 ms     9 ms  192.122.175.59
  8    18 ms     7 ms     7 ms  fourhundredge-0-0-0-65.4079.agg1.ashb.net.internet2.edu [163.253.1.147]
  9     *     3299 ms     8 ms  72.14.197.14
 10    10 ms     7 ms     7 ms  209.85.250.245
 11    10 ms     8 ms     7 ms  192.178.248.38
 12     9 ms     7 ms     7 ms  142.251.49.21
 13     *        *        *     Request timed out.
 14     6 ms     6 ms     6 ms  142.250.226.163
 15    11 ms     8 ms     7 ms  142.250.235.87
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        *     Request timed out.
 19     *        *        *     Request timed out.
 20     *        *        *     Request timed out.
 21     *        *        *     Request timed out.
 22     *        *        *     Request timed out.
 23     *        *        *     Request timed out.
 24     *        *        *     Request timed out.
 25     *        *        *     Request timed out.
 26     *        *        *     Request timed out.
 27     *        *        *     Request timed out.
 28    11 ms     9 ms     9 ms  wv-in-f91.1e100.net [142.251.163.91]

Trace complete.
```

```
C:\Users\dylxn>tracert stake.us

Tracing route to stake.us [172.64.155.194]
over a maximum of 30 hops:

  1     4 ms     4 ms     4 ms  10.246.191.254
  2     *        *        *     Request timed out.
  3     5 ms     4 ms     4 ms  ti-pe-default-eth2-11.ns.vcu.edu [172.16.4.25]
  4     4 ms     5 ms     4 ms  borderfw-wifi-out-3404.ns.vcu.edu [172.30.0.65]
  5     5 ms     6 ms     5 ms  border-rtr-tocb-hun1-250.ns.vcu.edu [192.168.250.2]
  6     9 ms     9 ms     9 ms  192.70.138.53
  7    12 ms    13 ms    10 ms  192.122.175.59
  8    10 ms     8 ms     8 ms  fourhundredge-0-0-0-64.4079.agg1.ashb.net.internet2.edu [163.253.1.145]
  9     9 ms     8 ms     8 ms  162.252.69.53
 10     8 ms     8 ms     8 ms  173.245.63.109
 11    11 ms     8 ms     9 ms  173.245.63.118
 12     9 ms     8 ms     8 ms  172.64.155.194

Trace complete.
```

- Your analysis of the hops taken for different destinations. (15 pt)

Both traces share the same early hops and exit through Internet2. YouTube took over twice the hops with heavy timeouts, routing into Google's own network. Stake.us was faster, timeout-free, and used Cloudflare.

Task 3: Learning About NETSTAT

Objective: To interpret network information from the NETSTAT utility and make informed assumptions about the network.

Instructions:

1. Utilize the NETSTAT utility to gather information about network connections on your computer.
2. Analyze and interpret the information displayed by NETSTAT.
3. Go beyond simply reporting what the utility displays and make informed assumptions about the state of your network based on the NETSTAT output.
4. Explain the significance of understanding the information provided by NETSTAT in terms of network security and troubleshooting.

   NETSTAT lets you see what your computer's actually connected to, so you can spot something suspicious and trace it back to the process behind it.

Lab Report Requirements

- Your NETSTAT findings and interpretations. (20 pt)

```
C:\Users\dylxn>netstat -ano

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       1652
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING       4
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING       5856
  TCP    0.0.0.0:27036          0.0.0.0:0              LISTENING       33944
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING       1328
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING       1156
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING       2376
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING       3868
  TCP    0.0.0.0:49672          0.0.0.0:0              LISTENING       5940
  TCP    0.0.0.0:49680          0.0.0.0:0              LISTENING       1276
  TCP    10.246.144.94:139      0.0.0.0:0              LISTENING       4
  TCP    10.246.144.94:49416    104.208.203.90:443     ESTABLISHED     6704
  TCP    10.246.144.94:50591    160.79.104.10:443      TIME_WAIT       0
  TCP    10.246.144.94:51303    172.217.119.4:443      ESTABLISHED     2820
  TCP    10.246.144.94:51304    172.217.119.4:443      ESTABLISHED     2820
  TCP    10.246.144.94:51610    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:52025    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:52026    172.217.117.4:443      ESTABLISHED     2820
  TCP    10.246.144.94:52027    172.217.117.4:443      ESTABLISHED     2820
  TCP    10.246.144.94:52470    142.251.179.188:5228   ESTABLISHED     4544
  TCP    10.246.144.94:53500    52.110.2.136:443       ESTABLISHED     5880
  TCP    10.246.144.94:54123    172.217.113.4:443      CLOSE_WAIT      2820
  TCP    10.246.144.94:54124    172.217.113.4:443      CLOSE_WAIT      2820
  TCP    10.246.144.94:54125    172.217.112.4:443      CLOSE_WAIT      2820
  TCP    10.246.144.94:54629    104.208.203.90:443     ESTABLISHED     1872
  TCP    10.246.144.94:55043    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:57195    52.110.2.136:443       ESTABLISHED     5880
  TCP    10.246.144.94:57584    104.18.36.252:443      ESTABLISHED     4544
  TCP    10.246.144.94:59608    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:59618    172.217.112.4:443      TIME_WAIT       0
  TCP    10.246.144.94:59865    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:62072    52.111.230.46:443      ESTABLISHED     20280
  TCP    10.246.144.94:62362    172.217.112.4:443      CLOSE_WAIT      2820
  TCP    10.246.144.94:62649    107.167.110.216:443    CLOSE_WAIT      4544
  TCP    10.246.144.94:63089    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:63495    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:63590    23.48.203.169:443      ESTABLISHED     4544
  TCP    10.246.144.94:64050    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:64171    52.96.87.194:443       ESTABLISHED     23752
  TCP    10.246.144.94:64172    52.96.87.194:443       ESTABLISHED     23752
  TCP    10.246.144.94:64504    162.254.199.165:443    ESTABLISHED     33944
  TCP    10.246.144.94:64649    128.172.1.5:53         TIME_WAIT       0
  TCP    10.246.144.94:65399    34.36.57.103:443       TIME_WAIT       0
  TCP    127.0.0.1:49702        127.0.0.1:49703        ESTABLISHED     1516
  TCP    127.0.0.1:49703        127.0.0.1:49702        ESTABLISHED     1516
  TCP    127.0.0.1:49704        127.0.0.1:49705        ESTABLISHED     1748
  TCP    127.0.0.1:49705        127.0.0.1:49704        ESTABLISHED     1748
  TCP    127.0.0.1:49706        127.0.0.1:49707        ESTABLISHED     6240
  TCP    127.0.0.1:49707        127.0.0.1:49706        ESTABLISHED     6240
  TCP    127.0.0.1:64485        0.0.0.0:0              LISTENING       33944
  TCP    127.0.0.1:64485        127.0.0.1:64493        ESTABLISHED     33944
  TCP    127.0.0.1:64487        0.0.0.0:0              LISTENING       33944
  TCP    127.0.0.1:64487        127.0.0.1:64492        ESTABLISHED     33944
  TCP    127.0.0.1:64492        127.0.0.1:64487        ESTABLISHED     12716
  TCP    127.0.0.1:64493        127.0.0.1:64485        ESTABLISHED     12716
  TCP    192.168.56.1:139       0.0.0.0:0              LISTENING       4
  TCP    [::]:135               [::]:0                 LISTENING       1652
  TCP    [::]:445               [::]:0                 LISTENING       4
  TCP    [::]:49664             [::]:0                 LISTENING       1328
  TCP    [::]:49665             [::]:0                 LISTENING       1156
  TCP    [::]:49668             [::]:0                 LISTENING       2376
  TCP    [::]:49669             [::]:0                 LISTENING       3868
  TCP    [::]:49672             [::]:0                 LISTENING       5940
  TCP    [::]:49680             [::]:0                 LISTENING       1276
  TCP    [::1]:7679             [::]:0                 LISTENING       2820
  TCP    [::1]:42050            [::]:0                 LISTENING       17956
  TCP    [::1]:49673            [::]:0                 LISTENING       6272
  UDP    0.0.0.0:123            *:*                                    5548
  UDP    0.0.0.0:5050           *:*                                    5856
  UDP    0.0.0.0:5353           *:*                                    4544
  UDP    0.0.0.0:5353           *:*                                    15776
  UDP    0.0.0.0:5353           *:*                                    13988
  UDP    0.0.0.0:5353           *:*                                    4544
```

```
  UDP    0.0.0.0:5353           *:*                                    13988
  UDP    0.0.0.0:5353           *:*                                    4544
  UDP    0.0.0.0:5353           *:*                                    4544
  UDP    0.0.0.0:5353           *:*                                    13988
  UDP    0.0.0.0:5353           *:*                                    15776
  UDP    0.0.0.0:5353           *:*                                    2252
  UDP    0.0.0.0:5353           *:*                                    4544
  UDP    0.0.0.0:5353           *:*                                    15776
  UDP    0.0.0.0:5353           *:*                                    4544
  UDP    0.0.0.0:5353           *:*                                    13988
  UDP    0.0.0.0:5353           *:*                                    4544
  UDP    0.0.0.0:5353           *:*                                    4544
  UDP    0.0.0.0:5353           *:*                                    13988
  UDP    0.0.0.0:5353           *:*                                    15776
  UDP    0.0.0.0:5353           *:*                                    4544
  UDP    0.0.0.0:5355           *:*                                    2252
  UDP    0.0.0.0:27036          *:*                                    33944
  UDP    0.0.0.0:51950          *:*                                    2252
  UDP    0.0.0.0:56272          160.79.104.10:443                      4544
  UDP    0.0.0.0:59222          160.79.104.10:443                      4544
  UDP    0.0.0.0:59648          *:*                                    2252
  UDP    10.246.144.94:137      *:*                                    4
  UDP    10.246.144.94:138      *:*                                    4
  UDP    10.246.144.94:1900     *:*                                    5792
  UDP    10.246.144.94:2177     *:*                                    1592
  UDP    10.246.144.94:27036    *:*                                    33944
  UDP    10.246.144.94:60497    *:*                                    5792
  UDP    127.0.0.1:1900         *:*                                    5792
  UDP    127.0.0.1:59669        127.0.0.1:59669                        6248
  UDP    127.0.0.1:60498        *:*                                    5792
  UDP    192.168.56.1:137       *:*                                    4
  UDP    192.168.56.1:138       *:*                                    4
  UDP    192.168.56.1:1900      *:*                                    5792
  UDP    192.168.56.1:2177      *:*                                    1592
  UDP    192.168.56.1:60496     *:*                                    5792
  UDP    [::]:123               *:*                                    5548
  UDP    [::]:5353              *:*                                    4544
  UDP    [::]:5353              *:*                                    4544
  UDP    [::]:5353              *:*                                    15776
  UDP    [::]:5353              *:*                                    13988
  UDP    [::]:5353              *:*                                    4544
```

```
  UDP    [::]:5353              *:*                                    4544
  UDP    [::]:5353              *:*                                    4544
  UDP    [::]:5353              *:*                                    15776
  UDP    [::]:5353              *:*                                    13988
  UDP    [::]:5353              *:*                                    4544
  UDP    [::]:5353              *:*                                    4544
  UDP    [::]:5353              *:*                                    15776
  UDP    [::]:5353              *:*                                    2252
  UDP    [::]:5353              *:*                                    13988
  UDP    [::]:5355              *:*                                    2252
  UDP    [::]:27036             *:*                                    33944
  UDP    [::]:51950             *:*                                    2252
  UDP    [::]:59648             *:*                                    2252
  UDP    [::1]:1900             *:*                                    5792
  UDP    [::1]:60495            *:*                                    5792
  UDP    [fe80::5ca9:b9ab:7b1b:8ec0%6]:546  *:*                                2388
  UDP    [fe80::5ca9:b9ab:7b1b:8ec0%6]:1900  *:*                               5792
  UDP    [fe80::5ca9:b9ab:7b1b:8ec0%6]:2177  *:*                               1592
  UDP    [fe80::5ca9:b9ab:7b1b:8ec0%6]:60494  *:*                              5792
  UDP    [fe80::e742:a335:4caf:15f8%18]:1900  *:*                              5792
  UDP    [fe80::e742:a335:4caf:15f8%18]:2177  *:*                              1592
  UDP    [fe80::e742:a335:4caf:15f8%18]:60493  *:*                             5792
```

- Your analysis of your network connection, based on the NETSTAT report. (20 pt)

Traffic looks like a normal workstation, not a server. Open ports suggest a managed/domain network (VCU lab). VirtualBox is likely installed (192.168.56.1). No unfamiliar IPs or ports no signs of malware.